

Hypernatremia — Diagnostic Algorithm





## 1. Assess volume

### WHAT YOU SEE

Bottom arrow: VOLUME → Uosm → DDAVP

### WHAT IT ENCODES

The diagnostic workup of hypernatremia follows three steps in order, shown by the bottom arrow: (1) assess VOLUME status (hypo-, eu-, or hypervolemic) to sort the broad mechanism; (2) measure URINE OSMOLALITY to ask whether the kidney is concentrating appropriately; (3) if DI is suspected, give DDAVP and recheck Uosm to split central from nephrogenic. Volume → Uosm → DDAVP is the spine of the whole algorithm.

### BOARD TRAP

WRONG answer: 'just measure urine sodium first to work up hypernatremia.' The first branch point is clinical VOLUME status, then urine osmolality; urine Na is a secondary clue within volume categories. Discriminator: follow the ordered sequence volume → Uosm → DDAVP rather than jumping to a single lab value.

## 2. Hypovolemic

### WHAT YOU SEE

Thin man labeled HYPOVOLEMIC, dry cracked lips, low cracked pulse

### WHAT IT ENCODES

Hypovolemic hypernatremia means water loss EXCEEDS sodium loss (net loss of hypotonic fluid). Exam shows volume depletion — dry cracked lips/mucosa, tachycardia, orthostasis, poor skin turgor. Causes: GI losses (osmotic diarrhea, vomiting, NG suction), osmotic diuresis (hyperglycemia, mannitol, urea), excessive sweating/burns, and renal losses including diuretics. Treatment restores volume with isotonic saline first, then free water.

### BOARD TRAP

WRONG answer: 'a hypovolemic hypernatremic patient has lost mostly salt and needs aggressive sodium repletion.' They have lost MORE water than salt (hypotonic loss) — they need volume resuscitation followed by FREE WATER, not added sodium. Discriminator: hypovolemic hyperNa = hypotonic fluid loss; replace volume (NS) then water, don't load salt.

## 3. Euvolemic

### WHAT YOU SEE

Monk figure labeled EUVOLEMIC

### WHAT IT ENCODES

Euvolemic hypernatremia reflects nearly PURE water loss with little change in total body sodium, so volume status looks roughly normal early on. The classic causes are diabetes insipidus (central or nephrogenic) and insensible/respiratory losses. Because sodium content is normal, these patients are not grossly volume-depleted unless they also cannot replace the water — the monk (calm, dry, no overt volume change) embodies the euvolemic picture.

### BOARD TRAP

WRONG answer: 'diabetes insipidus presents as a markedly hypovolemic, shocked patient.' DI is classically EUVOLEMIC (pure water loss) — patients compensate by drinking and only become hypernatremic/hypovolemic when access to water fails. Discriminator: euvolemic hyperNa with large dilute urine = DI; overt hypovolemia points more to GI/osmotic losses.

## 4. Hypervolemic

### WHAT YOU SEE

Door/figure HYPERVOLEMIC with Na GAIN, IV bag

### WHAT IT ENCODES

Hypervolemic hypernatremia is caused by SODIUM GAIN — total body sodium and volume both rise. Sources are almost entirely iatrogenic: hypertonic (3%) saline, repeated NaHCO<sub>3</sub> during resuscitation, hypertonic tube feeds, and hyperaldosteronism/Cushing (mild). Patients are volume-overloaded (edema, high urine Na). Management removes the source, gives free water (D5W), and may add a loop diuretic ± dialysis if renal function is impaired.

### BOARD TRAP

WRONG answer: 'hypervolemic hypernatremia is treated like other hypernatremias with isotonic saline.' Adding NaCl worsens the sodium load; the treatment is free water (D5W) ± a loop diuretic to remove excess sodium and volume. Discriminator: hypervolemic/edematous + high urine Na = sodium gain — remove sodium, give water, diurese; don't give saline.

## 5. Na GAIN

### WHAT YOU SEE

'Na GAIN' label with IV

### WHAT IT ENCODES

The 'Na GAIN' tag with an IV bag is the clue for the hypervolemic branch: an exogenous sodium load (hypertonic saline, NaHCO<sub>3</sub>, hypertonic feeds) raises both serum sodium and extracellular volume simultaneously. This is the rarest mechanism overall and is almost always iatrogenic in hospitalized patients. Urine sodium is HIGH because the kidney tries to excrete the excess sodium.

### BOARD TRAP

WRONG answer: 'a high urine sodium in hypernatremia always indicates the kidney is the source of water loss.' In sodium GAIN the urine Na is high because the kidney is appropriately dumping an exogenous sodium load — interpret urine Na together with volume status and urine osmolality. Discriminator: high urine Na + hypervolemia = sodium gain, not renal water wasting.

## 6. Urine osm gauge

### WHAT YOU SEE

Large gauge reading 800, marked >800 vs <800

### WHAT IT ENCODES

Urine osmolality is the pivotal lab. In a hyperosmolar patient a normal kidney should maximally concentrate urine (Uosm up to ~800–1200 mOsm/kg) to retain water. The gauge splits the algorithm: Uosm >700–800 means the kidney is responding appropriately (so the water loss is extrarenal, or there is sodium gain); a dilute Uosm <700–800 in the face of high serum osm is INAPPROPRIATE and points to renal water wasting (DI or osmotic diuresis).

### BOARD TRAP

WRONG answer: 'any concentrated-looking urine rules out a renal cause of hypernatremia.' Partial DI and osmotic diuresis can give intermediate Uosm; the key is whether Uosm is MAXIMAL/appropriate for the degree of hyperosmolality, and osmotic diuresis raises urine osm via solute while still wasting water. Discriminator: judge Uosm against the expected maximal concentration, and check for urinary solute (glucose/urea) to catch osmotic diuresis.



## 7. Uosm >800

### WHAT YOU SEE

Green check '>800' 

### WHAT IT ENCODES

A high urine osmolality (>700–800, green check) means the kidney's concentrating mechanism is intact and AVP is acting appropriately — so the body is correctly trying to conserve water. This directs you to EXTRARENAL water loss (GI, skin/sweat, insensible/respiratory) or to sodium gain. The concentrating defect of DI is excluded when the urine is maximally concentrated.

### BOARD TRAP

WRONG answer: 'a hypernatremic patient with maximally concentrated urine probably has diabetes insipidus.' Maximal Uosm EXCLUDES DI (the kidney is concentrating fine) and points to extrarenal water loss or sodium gain. Discriminator: high Uosm = appropriate renal response → look outside the kidney (GI, skin, lungs) or for a sodium load.

## 8. Uosm <800

### WHAT YOU SEE

Red X '<800' 

### WHAT IT ENCODES

An inappropriately DILUTE urine (Uosm <700–800, red X) in a patient who is hyperosmolar is pathologic — a healthy kidney should be concentrating maximally. This identifies RENAL water wasting and narrows to diabetes insipidus (failure of AVP signaling) or osmotic diuresis (glucose, urea, mannitol dragging water out). The next step is to look for urinary solute and then trial DDAVP.

### BOARD TRAP

WRONG answer: 'dilute urine in a hypernatremic patient is reassuring and just means good hydration.' Dilute urine when serum is hyperosmolar is the OPPOSITE of normal — it signals renal water wasting (DI or osmotic diuresis) and requires further workup. Discriminator: low Uosm + high serum osm = inappropriate → DI vs osmotic diuresis; proceed to DDAVP trial.

## 9. Extra-renal

### WHAT YOU SEE

Flask labeled EXTRA-RENAL

### WHAT IT ENCODES

Extrarenal water loss is the diagnosis when Uosm is HIGH (kidney concentrating maximally): the water is leaving by a non-renal route — diarrhea/vomiting/NG suction (GI), sweating/burns/fever (skin), and insensible/respiratory losses. Urine Na is typically low here because the volume-depleted kidney is avidly reabsorbing sodium too. Treatment replaces free water and addresses the source of loss.

### BOARD TRAP

WRONG answer: 'high urine osmolality with low urine sodium points to a renal cause of the hypernatremia.' That combination indicates the kidney is appropriately conserving both water and sodium against an EXTRARENAL loss. Discriminator: high Uosm + low urine Na = extrarenal water loss; the kidney is the hero, not the culprit.

## 10. Renal

### WHAT YOU SEE

Flask labeled RENAL

### WHAT IT ENCODES

Renal water loss is the diagnosis when Uosm is inappropriately LOW: the kidney itself is wasting water. The two categories are diabetes insipidus (defective AVP production or response → dilute urine) and osmotic diuresis (excess filtered solute — glucose in DKA/HHS, urea after high-protein feeds or recovering ATN, mannitol — that obligates water excretion). Distinguish them by checking urine for solute and by the DDAVP trial (DI responds depending on type; osmotic diuresis does not).

### BOARD TRAP

WRONG answer: 'inappropriately dilute urine in hypernatremia always means diabetes insipidus.' Osmotic diuresis also wastes renal water but is driven by excess urinary SOLUTE (glucose, urea, mannitol), not an AVP problem, and won't respond to DDAVP. Discriminator: measure urinary solute/glucose — high solute output = osmotic diuresis; truly free-water diuresis with low solute = DI.

## 11. DDAVP test

### WHAT YOU SEE

Man at lectern holding DDAVP vial

### WHAT IT ENCODES

Once renal water wasting points to DI, the desmopressin (DDAVP) trial separates the two types: administer DDAVP and re-measure urine osmolality. This works because central DI has intact kidneys that respond to supplied hormone, while nephrogenic DI is resistant downstream. In an already-hypernatremic patient this trial replaces the dangerous water-deprivation test.

### BOARD TRAP

WRONG answer: 'do a full overnight water-deprivation test before ever giving DDAVP in a hypernatremic DI patient.' If the patient is already hyperosmolar with dilute urine, go straight to the DDAVP trial — withholding water further is dangerous. Discriminator: hypernatremic + dilute urine → DDAVP trial directly; water deprivation is only for the normonatremic polyuric patient.

## 12. Central DI ↑Uosm

### WHAT YOU SEE

Up arrow ↑ over CENTRAL kidney

### WHAT IT ENCODES

Central DI: after DDAVP, urine osmolality RISES substantially — by >50% (often >100% in complete central DI) — because the kidney was always able to respond and merely lacked the hormone. This rise is the defining positive test for central DI and confirms the lesion is in AVP production/release, not the kidney. It also predicts a good therapeutic response to maintenance desmopressin.

### BOARD TRAP

WRONG answer: 'only a doubling or complete normalization of Uosm counts as a positive DDAVP response for central DI.' A rise of >50% in urine osmolality after DDAVP is sufficient to indicate central DI (partial central DI gives an intermediate but clear rise). Discriminator: any substantial (>50%) Uosm increase after DDAVP = central DI.



### 13. Nephrogenic no rise

#### WHAT YOU SEE

Minus sign over NEPHROGENIC kidney

#### WHAT IT ENCODES

Nephrogenic DI: after DDAVP there is LITTLE or NO rise in urine osmolality (<50%, often negligible) because the collecting duct is resistant to AVP — the V2 receptor or aquaporin-2 channel is the broken link, so adding hormone changes nothing. This flat response confirms the defect is renal and directs treatment to thiazide ± amiloride ± low-salt diet and removing the cause (lithium, hypercalcemia, hypokalemia).

#### BOARD TRAP

WRONG answer: 'a flat Uosm response to DDAVP means the test failed and should be repeated with central DI assumed.' A negligible rise IS the diagnostic result — it confirms NEPHROGENIC DI. Discriminator: no/blunted Uosm rise after DDAVP = nephrogenic (AVP-resistant); the failure to respond is the answer, not a failed test.

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### 14. No water = never fix

#### WHAT YOU SEE

Red X over running man / locked door

#### WHAT IT ENCODES

The overarching message (red X over the man at the locked door): whatever the mechanism, hypernatremia only PERSISTS when thirst is impaired or access to water is blocked, because an intact thirst-and-access system normally corrects water deficits quickly. So in every case you must identify and fix the water-access barrier — give water, treat adipsia, and address why the patient could not drink — or the hypernatremia recurs.

#### BOARD TRAP

WRONG answer: 'once you identify and treat the renal or extrarenal cause, the water-access status doesn't matter.' If the patient still cannot sense thirst or reach water, hypernatremia will recur regardless of the primary cause being treated. Discriminator: always answer 'why couldn't this patient drink enough water?' — the access/thirst barrier is the common final pathway in persistent hypernatremia.

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